

Introduction to Data Cleaning

Data cleaning is the process of detecting and correcting errors or inconsistencies in raw data to improve its quality and reliability. This crucial step ensures that your data is accurate, complete, and ready for analysis. Without effective data cleaning, any insights or models built from the data may be misleading or incorrect. You will often encounter several typical problems in raw datasets:

- Missing values: cells or entries where data is absent;
- Duplicates: repeated entries that can skew analysis;
- Inconsistencies: variations in how data is recorded, such as different date formats or inconsistent capitalization.

Understanding these issues is the first step toward producing trustworthy results from your data projects.

```
import pandas as pd

# Create a simple DataFrame with missing and duplicate values
data = {
    "Name": ["Alice", "Bob", "Charlie", "Bob", "Eve", None],
    "Age": [25, 30, 35, 30, None, 22]
}
df = pd.DataFrame(data)

print("Original DataFrame:")
print(df)

# Check for missing values
print("\nMissing values in each column:")
print(df.isnull().sum())

# Check for duplicate rows
print("\nDuplicate rows:")
print(df.duplicated())
```

When working with real-world data, you will often see **missing values** where information was not recorded, as well as **duplicate records** that can lead to overcounting. Identifying and addressing these issues is a core part of the data cleaning process.